

Cerebral and Muscle Oxygenation during Repeated Shuttle Run Sprints with Hypoventilation

Authors

Xavier Woorons^{1, 2}, Olivier Dupuy³, Patrick Mucci¹, Gregoire P. Millet⁴, Aurelien Pichon³

Affiliations

- 1 Univ. Lille, Univ. Artois, Univ. Littoral Côte d'Opale, EA 7369 – URePSSS – Unité de Recherche Pluridisciplinaire Sport Santé Société, F-59000 Lille, France
- 2 ARPEH, Association pour la Recherche et la Promotion de l'Entraînement en Hypoventilation, 18 Rue Saint Gabriel 59800 Lille, France
- 3 Université de Poitiers. Faculté des Sciences du Sport, Laboratoire Move, EA 6314, Poitiers, France
- 4 ISSUL, Institute of Sport Sciences, University of Lausanne, Building Synathlon, Campus Dorigny 1015 Lausanne, Switzerland

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Correspondence

Dr. Xavier Woorons, PhD

Faculté des Sciences du Sport

Unité de Recherche Pluridisciplinaire Sport Santé Société

(URePSSS)

9 rue de l'Université

59790 Ronchin

France

Tel.: +33/661/845 139, Fax: +33/3/20 88 73 63

xavier.woorons@gmail.com

ABSTRACT

Ten highly-trained Jiu-Jitsu fighters performed 2 repeated-sprint sessions, each including 2 sets of 8 x ~6 s back-and-forth running sprints on a tatami. One session was carried out with normal breathing (RSN) and the other with voluntary hypoventilation at low lung volume (RSH-VHL). Prefrontal and vastus lateralis muscle oxyhemoglobin ([O₂Hb]) and deoxyhemoglobin ([HHb]) were monitored by near-infrared spectroscopy. Arterial oxygen saturation (SpO₂), heart rate (HR), gas exchange and maximal blood lactate concentration ([La]_{max}) were also assessed. SpO₂ was significantly lower in RSH-VHL than in RSN whereas there was no difference in HR. Muscle oxygenation was not different between conditions during the entire exercise. On the other hand, in RSH-VHL, cerebral oxygenation was significantly lower than in RSN (-6.1 ± 5.4 vs $-1.5 \pm 6.6 \mu\text{m}$). Oxygen uptake was also higher during the recovery periods whereas [La]_{max} tended to be lower in RSH-VHL. The time of the sprints was not different between conditions. This study shows that repeated shuttle-run sprints with VHL has a limited impact on muscle deoxygenation but induces a greater fall in cerebral oxygenation compared with normal breathing conditions. Despite this phenomenon, performance is not impaired, probably because of a higher oxygen uptake during the recovery periods following sprints.

Introduction

Intermittent sports (e. g., team, racket or combat sports), as opposed to continuous-cyclic sports (e. g., cycling, running or swimming), are characterized by the repetition of maximal or near-maximal efforts of short duration (generally < 30 s) interspersed with brief recovery periods (< 60 s). In these sports, the objective is to

maintain the highest intensity during efforts to increase the effectiveness of the players/fighters during matches or combat. This ability has been termed repeated sprint ability (RSA) [18].

Over recent years, it has been shown that performing a repeated sprint training under normobaric hypoxic conditions (RSH) could be an effective approach to improving RSA. The majority of studies that investigated the effects of RSH have found larger perfor-

mance gains in these conditions than with the same training performed in normoxia [6]. It is noteworthy that 2 recent repeated-sprint studies have reported that when the hypoxic conditions were created through voluntary hypoventilation at low lung volume (VHL), the improvement in RSA performance was larger than in normoxia [16, 31]. Since 2007, it has clearly been demonstrated that VHL could induce severe arterial oxygen desaturation [37–40] leading to muscle deoxygenation [36, 40]. In these conditions, the hypoxic state has been shown to be similar to what is observed at an altitude of about 2400 m [35]. Even though this level is lower than what is commonly used for RSH (i. e., 3000–3500 m), it may induce beneficial physiological adaptations to performance [38, 41]. The VHL approach has thus been recognized as one of the altitude training methods available and useful for a wide range of sporting activities [17].

Of interest is that during repeated-sprint exercise (RSE), the performance decrement (velocity or power output of the sprints) is not larger with VHL than with normal breathing conditions [16, 31, 40]. This could be due to the fact that the levels of oxygen consumption are increased during the recovery periods following exercise with VHL [40]. Such phenomenon may lead to greater phosphocreatine resynthesis and/or lactate oxidation thus allowing the maintenance in performance. An elevated contribution of anaerobic glycolysis to counteract the drop in SpO_2 induced by VHL [36, 37] may also prevent impairment in performance. These outcomes are of importance since the efficiency of RSH relies on the repetition of maximal intensity efforts in order to recruit and induce physiological adaptation in fast-twitch fibers [11, 25].

The role of a lower cerebral oxygenation has been used to explain the greater decrement of repeated-sprint performance under hypoxic conditions [28]. A fall in cerebral oxygenation when exercising in severe hypoxia (arterial oxygen saturation (SpO_2) below 75 %; altitude > 3000 m) could have a negative impact on the central command, which could lead to central fatigue [13] and consequently to a lower muscle activation and performance [33]. Whether the same phenomenon could occur during RSH induced by VHL is currently unknown.

So far, no study has ever investigated the effects of VHL exercise on cerebral oxygenation. Furthermore, no acute physiological data are available for repeated-maximal efforts performed with VHL in running, or more generally in upright motions, which are the main exercise mode in team, racket or combat sports. More than straight-line motions, the practice of these sports involves numerous changes in direction. Yet, this latter feature requires discontinuous eccentric and concentric muscle activations [8]. Consequently, the physiological responses may be different from straight line motions. The main goal of the present study was therefore to determine the acute effects of repeated shuttle-run sprints with VHL on cerebral and muscle oxygenation. We hypothesized that both a greater muscle and cerebral deoxygenation would occur under these conditions. However, we also assumed that this may be compensated by an increase in the anaerobic glycolysis and/or a higher oxygen consumption during the recovery periods, which could prevent an impairment in performance.

Materials and Methods

Subjects

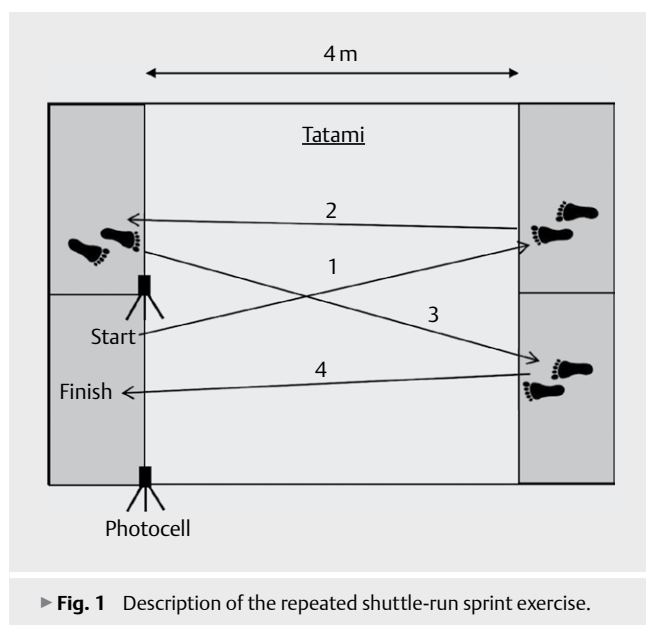
Ten highly-trained Jiu-Jitsu fighters (7 men and 3 women), with about 10–12 years of practice, were selected to participate in this study. Their physical characteristics (mean $\pm$ SD) were age 19.2 ± 2.3 years, height 171.2 ± 8.1 cm and body mass 68.2 ± 10.6 kg. Two of the participants (one male and one female) were under 21 (U21) world champions, 3 others were ranked in the top 5 European fighters (U18 and above 21) and the last 4 had a national or regional level. The training volume of the participants was 7 sessions per week on average for a total of approximately 12 h. Jiu-Jitsu fighters were chosen since this combat sport requires the repetition of high-intensity bouts interspersed with short periods of recovery. Furthermore, the use of repeated sprint training with RSH-VHL was thought as potentially beneficial for combat sports in general. All participants were non-smokers, sea-level residents and not exposed to altitude (i. e., > 500 m above sea level) in the few months preceding the experiment. They gave their written informed consent to participate in this study, which was approved by the Ethics Committee for the Research in Sports Science (CERSTAPS) and complied with the Declaration of Helsinki (2008). Furthermore, the present study meets the ethical standards of the International Journal of Sports Medicine [22].

Study design

The protocol consisted of performing 2 testing sessions of repeated shuttle-run sprints separated by 24 h. Each testing session comprised 2 sets of $8 \times \sim 6$ s back-and-forth sprints at maximal (all-out) velocity. Between repetitions and sets, subjects observed a passive recovery (standing position) of 15 s and 3 min respectively. In a randomized and counterbalanced order, each participant performed one of the repeated-sprint sessions with normal breathing (RSN) and the other in hypoxemia induced by VHL (RSH-VHL). This technique has been used and fully described in a recent repeated-sprint study [40]. Briefly, during RSH-VHL, the subjects were told to do a normal exhalation at the start of each shuttle-run sprint and then to hold their breath until its completion.

Shuttle-run sprints were carried out on a tatami (barefoot) between colored mats separated by a distance of 4 m. The course is displayed in ► **Fig. 1**. After the start was given from one colored mat, about 0.5 m behind photocells (on the left corner), subjects had to make a first back-and-forth sprint which ended next to the left photocell. Then, without pausing, they had to make a second back-and-forth and complete the course by passing through the photocells. They were told to put both feet on the colored mat before turning 180° and sprinting to the other mat. This testing procedure was implemented in collaboration with Jiu-Jitsu national coaches in order to have a standardized and reliable test which takes into account the specificity of this sport discipline. Within the month preceding the experiment, all subjects participated at least in 2 sessions to familiarize with the testing procedures as well as the VHL technique.

Before starting the first set, subjects performed a thorough warm-up of about 15-min. The warm-up included low- and moderate-intensity running around the gym, active mobilization, dy-



dynamic stretching, running drills and 3–4 back-and-forth sprints over the same course as during the test. The warm-up was followed by a single shuttle-sprint at maximal intensity in order to get the reference time. The test began after a 10-min period of rest. At the first shuttle-run sprint of set 1 (S1), subjects were required not to exceed 5% of the reference time. If they failed to do so, they had to restart the set after a 5-min period of rest.

Measurements

Performance

Time of the shuttle sprints was measured with photocells whose height was adjusted according to the height of the participants' hip. Time of each sprint as well as the mean and best time of each of the 2 sets were analyzed. Furthermore, the fatigue index was assessed by calculating the percentage decrement score as follows:

Fatigue index = $(100 \times (\text{total shuttle-sprint time of the set} / \text{ideal shuttle-sprint time of the set})) - 100$

Where:

total shuttle-sprint time = sum of the time of all sprints of the set.
ideal sprint time = number of sprints (i. e., 8) $\times$ best sprint time of the set.

This formula has been found to be the most valid and reliable method for quantifying fatigue in tests of multiple-sprint performance [19].

Gas exchange

Gas exchange was continuously recorded during the entire repeated shuttle-run sprints through a breath-by-breath portable system (K4b², Cosmed, Rome, Italy). The standardized calibration procedures were performed before the beginning of exercise as recommended by the manufacturer. These included air calibration, turbine calibration with a standard 3000-mL syringe, gas calibration with a certified commercial gas preparation (oxygen: 16%, carbon dioxide: 5%) and delay calibration to ensure accurate readings dur-

ing the testing and to check the alignment between the gas flow and gas concentrations. The breath-by-breath measurements included tidal volume, breathing frequency, expired ventilation, oxygen uptake ($\dot{V}O_2$), carbon dioxide production, end-tidal oxygen pressure and carbon dioxide pressure (PETCO₂). The ventilatory equivalent for both oxygen and carbon dioxide were calculated. Since gas exchange can obviously not be measured during sprints with breath holdings, data were analyzed over the 15 s of the recovery periods. Nevertheless, for a better understanding of the outcomes and conclusions of the present study, we also chose to analyze and present data of tidal volume, breathing frequency, expired ventilation and $\dot{V}O_2$ measured during the 6-sec sprints in RSN.

Heart rate

Heart rate (HR) was continuously recorded using the K4b² device (Cosmed, Rome, Italy) presented above, associated with a Polar T34 strap (Polar Electro Inc, Lake Success, NY, USA) which was placed on the chest of the participants. Mean HR was analyzed over the 15 s of the recovery periods (for the same reasons as indicated above) and the oxygen pulse was calculated. We also analyzed the peak HR for each of the 2 sets.

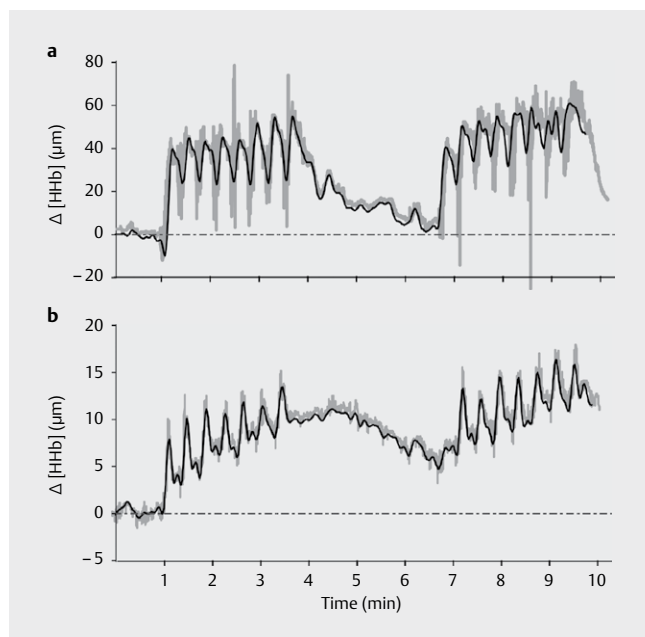
Arterial oxygen saturation

To ensure accurate and reliable measurements of arterial oxygen saturation (SpO₂), we chose to use the pulse oximeter Nellcor N-595 (Pleasanton, CA, USA) with the adhesive forehead sensor Max-Fast (Nellcor, Pleasanton). A portable oximeter with digit sensor may have led to wrong measurements in such testing conditions. SpO₂ was continuously measured during S1 and set 2 (S2) and recorded at a frequency of 0.5 hz. The measurements were performed by an operator who followed the subjects during their back-and-forth sprints using the special rod supplied with the Aquatrainer module (Cosmed, Rome, Italy), which is utilized for collecting gas exchange data with the K4b² in swimming. The sensor was applied above the right orbital area and covered by an adjustable headband to ensure gentle, consistent pressure. This kind of sensor has already been used during extreme conditions of exercise (i. e., swimming at high intensity) [31, 37, 41] and has been shown to provide accurate and reliable estimations of arterial oxygen saturation [15]. After the recordings, data were collected using a data acquisition system (Score Analysis Software, Nellcor, Pleasanton). They were then averaged and analyzed over 6 s, which corresponded approximately to the duration of each sprint. During RSH-VHL, as previously reported [40], SpO₂ begins to drop at the end or just after the 6-sec sprints. For these reasons, independently of sprint/recovery periods, the results are presented for the 6-sec periods in which the minimum values of SpO₂ were reached.

Near-infrared spectroscopy

Muscle and cerebral oxygenation were assessed using a near-infrared spectroscopy (NIRS) technique [5], at wavelengths between 760 and 850 nm. For muscle oxygenation, we used the PortaMon device (Artinis, Zetten, The Netherlands), which was placed at the lower third of the right-leg vastus lateralis muscle (the dominant (pivot) leg used for the direction changes in the course of the trial), parallel to the long axis of the muscle and with an inter-optode spacing of 40 mm. The vastus lateralis has been shown to be mainly involved in

running sprints including direction changes [8]. The probe was attached to the skin with double-sided tape and firmly fastened with an opaque cotton elastic band wrapped around subjects' thigh. Position of the probe was marked at the first testing session with a permanent pen for accurate repositioning at the second testing session. For cerebral oxygenation, we used the Portalite device (Artinis, Zetten, The Netherlands), which was positioned on the surface of the left prefrontal cortex (above the eyebrow, between the midline of the skull and the temporalis muscle), as reported in previous studies on repeated sprints [4, 28, 34]. The probe was attached with double-sided tape and a head wrap was added to create a dark environment and maintain a stable position. In the absence of a clear standard value for human quadriceps muscle during such a specific trial, we used a standard differential pathlength factor of 4.0 for the vastus lateralis muscle. For the prefrontal cortex, an age-dependent differential pathlength factors was used [10]. All signals were recorded with a sampling frequency of 10 Hz. For the vastus lateralis muscle, a 10th-order low-pass zero-phase Butterworth filter (cut-off frequency 0.1 Hz) [12] was applied to reduce artefacts and smooth perturbations in the signal provoked by the back-and-forth runs. For the prefrontal data, after carefully checking and testing different treatment methods, it appeared that a low-pass filter with a cut-off frequency of 0.1 Hz was also the best procedure to reduce artefacts and smooth the signal. ► **Fig. 2** provides data of cerebral and muscle $\Delta[\text{HHb}/\text{Mb}]$ from a representative subject before and after treatment. Concentrations of oxyhemoglobin ($[\text{O}_2\text{Hb}]$) and deoxyhemoglobin ($[\text{HHb}]$) were recorded. To analyze muscle and cerebral oxygenation, we used an "oxygenation index," which was defined by the



► **Fig. 2** Representative data of the change in muscle **a** and cerebral **b** deoxyhemoglobin ($\Delta[\text{HHb}]$) from a single subject during the repeated shuttle-run exercise, illustrating the effects of a 10th-order low-pass zero-phase Butterworth filter with cut-off frequency of 0.1 Hz compared with raw data. The grey areas represent raw $[\text{HHb}]$ data. Black lines are the resulting data after the application of the Butterworth filter.

difference between $[\text{O}_2\text{Hb}]$ and $[\text{HHb}]$ ($[\text{Hb}_{\text{diff}}]$). This index has been used in previous studies [21, 24, 36] and considered reliable for assessing muscle oxygenation. Furthermore, $[\text{Hb}_{\text{diff}}]$ has been shown to be a relevant alternative to $[\text{HHb}]$ when total hemoglobin is not constant [32]. We also calculated total Hb ($[\text{THb}]$) by summing $[\text{O}_2\text{Hb}]$ and $[\text{HHb}]$ to get an index of the regional blood volume. For the 2 sets, the change (Δ) in $[\text{O}_2\text{Hb}]$, $[\text{HHb}]$, $[\text{THb}]$ and $[\text{Hb}_{\text{diff}}]$ were assessed from the resting values recorded over the 2 min preceding the start of the test. The measurements were therefore normalized from these recordings (arbitrarily defined as 0 μm). During repeated sprint exercise, muscle $[\text{HHb}]$ generally reaches its maximum value at the end of the sprints whereas, in the same time, the minimum values of muscle $[\text{O}_2\text{Hb}]$ and $[\text{THb}]$ are recorded [12]. On the other hand, cerebral $[\text{HHb}]$, $[\text{O}_2\text{Hb}]$ and $[\text{THb}]$ increase during sprints due to cerebral vasodilation [28]. For these reasons, Peak $[\text{HHb}]$ and both nadirs $[\text{O}_2\text{Hb}]$ and $[\text{THb}]$ of the vastus lateralis were analyzed for each sprint and for each subject whereas for the prefrontal cortex, only peak values were analyzed. Since during RSH-VHL it is possible that the peak and nadir values are reached in the few seconds following the end of the sprint [40], we adopted a rolling approach, as shown previously [12, 23]. A sliding window averaging data over a 4-sec period was thus used for all NIRS variables. It has recently been demonstrated that the use of a digital filter to smooth NIRS data combined with a rolling approach was the best method to determine peaks and nadirs for accurate interpretation of muscle oxygenation trends during repeated sprint exercise [27].

Rating of perceived exertion (RPE) and blood lactate concentration $[\text{La}]$

In both exercise conditions, just after the end of the second set, RPE was assessed for legs and breathing using the Borg scale (range 0–10). Furthermore, a blood sample was taken from the earlobe of the subjects at the second and third minute following the end of the second set to obtain $[\text{La}]$ (Lactate Pro, Arkray, Japan). The highest value of the 2 samples ($[\text{La}]_{\text{max}}$) was kept for the analyses.

Statistical analysis

All the results are expressed as mean $\pm$ SD. After testing data for distribution normality and variance homogeneity, we performed 2-way analyses of variance (ANOVA) for repeated measures to determine whether there was a difference in performance and in NIRS data between exercise conditions for each sprint of both sets. Two-way ANOVA for repeated measures were also performed to compare mean values (performance, gas exchange, cerebral $\Delta[\text{Hb}_{\text{diff}}]$, SpO_2 and HR), the minimum SpO_2 , peak HR and the fatigue index in each of the 2 sets. When a significant interaction effect was found, the Bonferroni post-hoc test was carried out to localize the differences. We used paired Student T-test to assess the differences in RPE and $[\text{La}]_{\text{max}}$ between conditions and to evaluate the differences in muscle and breathing RPE within conditions. The relationships between the change in the oxygen pulse, $\Delta\dot{V}\text{O}_2$, the mean SpO_2 , PETCO_2 and cerebral $\Delta[\text{Hb}_{\text{diff}}]$ were analyzed in S2 of RSH-VHL by performing Person linear regressions. All analyses were made using Sigmapstat 4.0 software (Systat Software, CA, USA). Null hypothesis was rejected at $p < 0.05$.

Results

Performance

Running performance (i. e., time of each sprint) was not different between conditions throughout exercise (► Fig. 3). There was no difference between RSH-VHL and RSN in both the best (S1: 5.83 ± 0.3 vs 5.70 ± 0.3 s; S2: 5.73 ± 0.2 vs 5.65 ± 0.3 s; $p = 0.3$) and the mean time of each set (S1: 5.99 ± 0.3 vs 5.92 ± 0.3 s; S2: 5.99 ± 0.3 vs 5.90 ± 0.3 s; $p = 0.41$). The best ($p = 0.3$) and the mean time ($p = 0.83$) of each set were not different between S1 and S2 both in RSH-VHL and in RSN. We found an interaction effect for the fatigue index ($p = 0.02$), which was lower in RSH-VHL than in RSN at S1 (2.8 ± 1.0 vs 3.96 ± 1.1 %; $p = 0.03$) but not different between conditions at S2 (4.5 ± 2.0 vs 4.5 ± 1.4 %; $p = 0.98$). The fatigue index was lower at S1 than at S2 in RSH-VHL ($p = 0.01$) and not different between sets in RSN ($p = 0.37$).

Arterial oxygen saturation

The mean SpO_2 reached at each repetition was lower in RSH-VHL than in RSN both at S1 (92.3 ± 2.2 vs 97.6 ± 1.0 %; $p < 0.01$) and S2 (89.8 ± 3.2 vs 97.7 ± 1.2 %; $p < 0.01$). The minimum SpO_2 was

also significantly lower in RSH-VHL than in RSN (► Fig. 4a). The mean SpO_2 during RSH-VHL in S2 was significantly correlated to the change in the oxygen pulse ($R = -0.66$, $p = 0.04$) but not to $\Delta \dot{V}\text{O}_2$ ($R = -0.54$, $p = 0.1$).

Gas exchange and heart rate

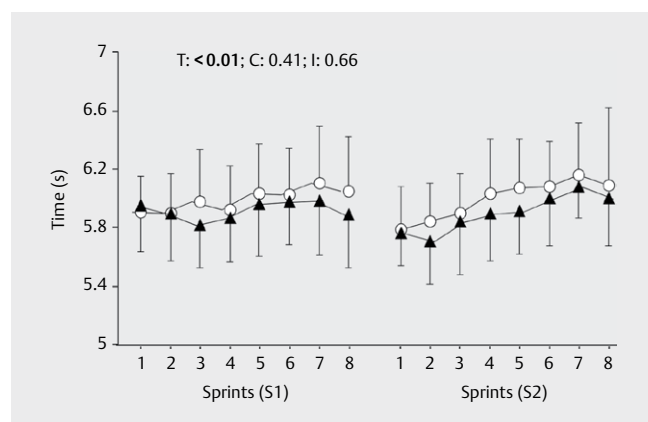
On average, during the 15 s of the recovery periods following sprints, $\dot{V}\text{O}_2$, the oxygen pulse, tidal volume and PETCO_2 were higher, and breathing frequency, expired ventilation, the ventilatory equivalent for oxygen and carbon dioxide and end-tidal oxygen pressure were lower in RSH-VHL than in RSN (► Table 1). On the other hand, there was no difference in carbon dioxide production, mean HR and peak HR (► Fig. 4b) between conditions. During sprints in RSN, tidal volume (S1: 2.11 ± 0.4 L; S2: 2.21 ± 0.4 L) and $\dot{V}\text{O}_2$ (S1: 37.9 ± 5.1 mL.min⁻¹.kg⁻¹; S2: 38.2 ± 6.6 mL.min⁻¹.kg⁻¹) were lower than during the recovery periods in RSH-VHL ($p < 0.01$) whereas breathing frequency (S1: 48.1 ± 13.4 b.min⁻¹; S2: 49.5 ± 9.1 b.min⁻¹) and expired ventilation (S1: 101.4 ± 21.0 L.min⁻¹; S2: 109.1 ± 21.6 L.min⁻¹) were not different ($p = 0.56$ and 0.14). Furthermore, breathing frequency ($p < 0.01$), expired ventilation ($p < 0.01$) and ($p < 0.05$) were higher during the recovery periods following RSN than during the sprint periods. On the other hand, there was no difference in tidal volume ($p = 0.93$) between both periods.

NIRS measurements

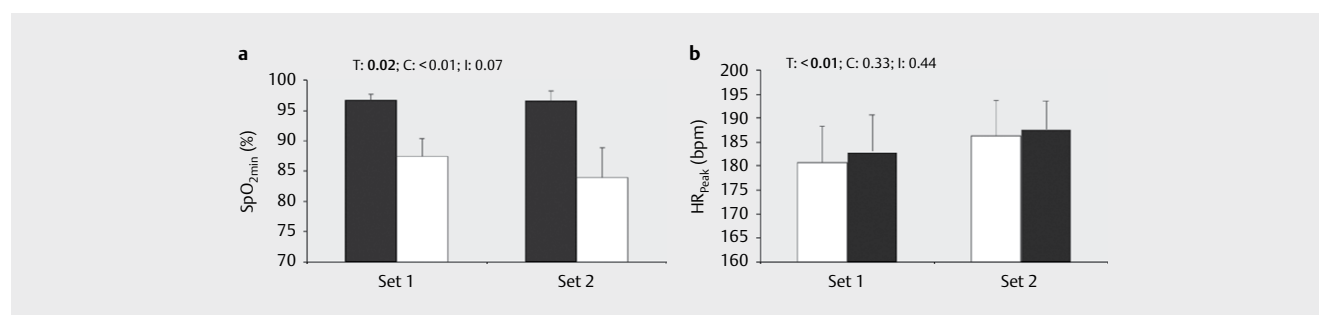
► Fig. 5 displays the mean concentration changes in cerebral and muscle NIRS signals across repeated shuttle-sprints in RSH-VHL and RSN.

Cerebral data

The ANOVA revealed that there was no difference between conditions in $\Delta[\text{O}_2\text{Hb}]$ and $\Delta[\text{THb}]$. On the other hand, $\Delta[\text{HHb}]$ was higher and $\Delta[\text{Hb}_{\text{diff}}]$ lower in RSH-VHL than in RSN. The mean $\Delta[\text{Hb}_{\text{diff}}]$ was also lower in RSH-VHL than in RSN both in S1 (-6.3 ± 4.4 vs -1.9 ± 5.0 μm , $p = 0.04$) and S2 (-5.8 ± 4.5 vs -1.2 ± 7.6 μm , $p = 0.04$). Compared to baseline values, there was an increase in both $[\text{HHb}]$ and $[\text{THb}]$ and a decrease in $[\text{Hb}_{\text{diff}}]$. Conversely, no significant change occurred for $[\text{O}_2\text{Hb}]$. In S2 of RSH-VHL, the mean $\Delta[\text{Hb}_{\text{diff}}]$ was significantly correlated to the mean SpO_2 ($R = 0.87$, $p < 0.001$) and tended to be correlated to PETCO_2 ($R = 0.55$, $P = 0.1$).



► Fig. 3 Time achieved at each sprint of the first (S1) and the second set (S2) of the repeated shuttle-run sprint exercise performed with voluntary hypoventilation at low lung volume (open circles) and with normal breathing (filled triangles). T, time effect; C, condition effect; I, interaction effect ($T \times C$).



► Fig. 4 Peak heart rate (HR_{peak}) a and minimum arterial oxygen saturation ($\text{SpO}_{2\text{mean}}$) b measured in set 1 and set 2 during the repeated shuttle-run sprints performed with voluntary hypoventilation at low lung volume (RSH-VHL, white bars) and with normal breathing (RSN, black bars). T, time effect; C, condition effect; I, interaction effect ($T \times C$).

► **Table 1** Mean cardiorespiratory values during the recovery periods in S1 and S2.

	RSH-VHL		RSN		ANOVA p value		
	S1	S2	S1	S2	T	C	T x C
Vt (L)	2.50±0.6	2.43±0.6	2.14±0.4	2.18±0.5	0.47	<0.01	0.16
Bf (breaths.min ⁻¹)	44.3±8.6	48.9±6.6	61.1±13.4	63.2±8.3	0.03	<0.01	0.50
VE (L.min ⁻¹)	110.8±32	118.2±28	130.8±27	137.9±25†	<0.01	<0.01	0.99
VO ₂ (mL.min ⁻¹ .kg ⁻¹)	44.5±6.6	46.5±6.5	43.6±6.0	42.7±7.4	0.60	0.03	0.11
VO ₂ (L.min ⁻¹)	3.02±0.6	3.15±0.6	2.97±0.6	2.91±0.6	0.64	0.03	0.12
HR _{mean} (bpm)	171.0±7.8	176.5±6.9	173.5±6.1	179.8±5.7	<0.01	0.21	0.13
VO ₂ /HR (mL.beat ⁻¹)	17.7±3.7	17.9±3.2	17.2±3.9	16.2±3.9	0.41	<0.01	0.14
VCO ₂ (L.min ⁻¹)	3.02±0.6	2.86±0.6	3.04±0.7	2.86±0.6	0.06	0.90	0.98
VE/VO ₂ (L.min ⁻¹)	36.4±6.5	37.3±4.1	44.0±8.1	47.7±6.6	0.03	<0.01	0.20
VE/VCO ₂ (L.min ⁻¹)	36.6±6.6	41.6±5.7	43.7±11.2	48.5±8.6	<0.01	0.02	0.74
PET _O ₂ (mmHg)	109.9±3.6	112.6±3.1	114.6±3.6	117.7±2.9	<0.01	<0.01	0.69
PET _{CO} ₂ (mmHg)	38.8±5.1	34.4±2.9	35.2±3.5	30.9±2.7	<0.01	0.03	0.94

Values are mean ± SD; RSH-VHL, repeated sprints in hypoxaemia induced by voluntary hypoventilation at low lung volume; RSN, repeated sprints with normal breathing; S1, first set; S2, second set; T, time effect; C, condition effect; T x C, interaction effect (time x condition); Vt, tidal volume; Bf, breathing frequency; VE, expired ventilation; VO₂, oxygen uptake; HR, heart rate; VO₂/HR, oxygen pulse; VCO₂, carbon dioxide production; VE/VO₂, ventilatory equivalent for oxygen; VE/VCO₂, ventilatory equivalent for carbon dioxide; PET_{CO}₂, end-tidal carbon dioxide pressure; PET_O₂ end-tidal oxygen pressure; p<0.05.

Muscle data

We did not find any difference between conditions in Δ[O₂Hb], Δ[HHb], Δ[THb] and Δ[Hb_{diff}] during the entire exercise. Compared to normalized resting values, there was a decrease in [O₂Hb], [THb] and [Hb_{diff}] and an increase in [HHb] in both conditions.

Rating of perceived exertion and [La]_{max}

There was no difference in either breathing (8.8±1.1 vs 9.3±0.6; p=0.19) or muscle RPE (6.9±1.1 vs 6.8±0.7; p=0.7) between RSN and RSH-VHL. Muscle RPE was lower than breathing RPE in the 2 exercise conditions (p<0.01). Blood maximal concentration tended to be lower in RSH-VHL than in RSN (8.7±2.3 vs 10.5±1.9 mmol.L⁻¹; p=0.06).

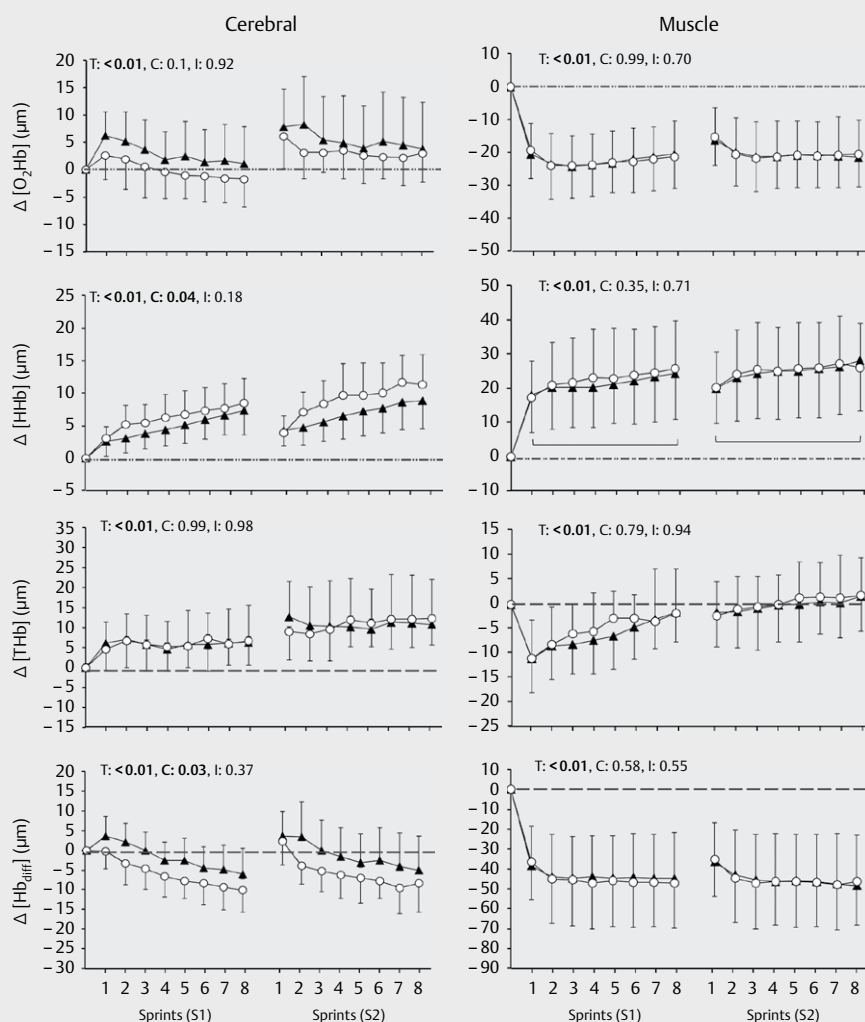
Discussion

This study is the first to investigate the acute effects of repeated shuttle-run sprints in hypoxemia induced by VHL. The major and original finding is that in RSH-VHL, cerebral oxygenation dropped below baseline values and was lower than in RSN throughout exercise. On the other hand, muscle deoxygenation was not different between exercise conditions in all of the repeated-shuttle sprints. Another interesting result is the fact that despite the large decrease in arterial and cerebral oxygenation, performance was maintained in RSH-VHL compared with RSN.

The minimum average level of SpO₂ recorded during the repeated-shuttle sprints with VHL (i. e., 83.9%, ► Fig. 4) confirms what has been reported since 2007, that is this breathing technique induces large arterial desaturation leading to severe hypoxemia [36, 37, 39]. This minimum level, as well as the average mean SpO₂ reached during RSH-VHL, are in line with findings of recent repeated-sprint studies with VHL conducted in swimming [31], cycling [40] and running [16]. Noteworthy, mean end-exercise SpO₂ levels of 82–85 % have been recorded during RSH in severe normobaric conditions (i. e., 3000 m, inspired fraction of oxygen (FiO₂) ~ 14 %

[4, 20]. To date, on the basis of the data available in literature, it appears that VHL is the only way to obtain large arterial deoxygenation during very short periods of exercise while remaining in normoxic environmental conditions. This makes this technique a good substitute for performing RSH without going to altitude or using devices to create hypoxia. VHL-induced fast and severe hypoxemia is made possible through both a wider alveolar-to-arterial O₂ partial pressure difference as a consequence of a greater ventilation-to-perfusion ratio inequality, and a right shift of the O₂ dissociation curve, under the effect of hypercapnia-induced acidosis [39]. This phenomenon is a specific response to breath holding at low lung volume, and is different from what can occur in acute hypoxia which rather leads to hypocapnia and in which the alveolar-to-arterial O₂ partial pressure difference is mainly induced by diffusion limitation rather than a greater ventilation-to-perfusion inequality [30].

Although SpO₂ dropped to low levels during RSH-VHL, this had no impact on muscle oxygenation. Both Δ[HHb] and Δ[Hb_{diff}] were not different between conditions throughout exercise whereas [THb] remained mostly unchanged compared to baseline in both RSH-VHL and RSN. This result was unexpected since we recently reported greater muscle deoxygenation (i. e., higher [HHb] and lower [O₂Hb]) during the second part of a cycling RSE performed with VHL compared with the same exercise in normal breathing conditions [40]. This discrepancy may be partly explained by the use of direction changes during the type of RSE implemented in the present study. This kind of exercise indeed requires eccentric and concentric muscle efforts due to deceleration-acceleration actions [2] and a discontinuous muscle activation because of the 180° turns. This may reduce the load of the vastus lateralis muscle, as suggested before [8], which may have an influence on the NIRS signals. Another possible explanation is that Jiu-Jitsu fighters, compared to endurance athletes, could be limited in their capacity to increase muscle O₂ extraction (i. e., higher [HHb] [21]) to compensate for the reduced O₂ supply under hypoxic conditions. It is, however, interesting to note that



► **Fig. 5** Cerebral and muscle concentration changes (Δ) in oxyhemoglobin ($[O_2Hb]$), deoxyhemoglobin ($[HHb]$), total hemoglobin ($[THb]$) and $[Hb_{diff}]$ measured by near infrared-spectroscopy (NIRS) at each sprint of set 1 **S1** and set 2 **S2** of the repeated shuttle-run sprint exercise performed with voluntary hypoventilation at low lung volume (RSH-VHL, open circles) and with normal breathing (RSN, filled triangles). T, time effect; C, condition effect; I, interaction effect ($T \times C$). Changes are expressed in arbitrary units (μm) from resting baseline set to 0 μm .

even when performed at a high degree of hypoxia, RSE does not necessarily provoke larger muscle deoxygenation than the same exercise performed in normoxic conditions [4, 28].

Unlike what happened at the muscle level, the shuttle sprints with VHL had a significant effect on cerebral oxygenation. In this condition, the oxygenation index of the prefrontal cortex (i. e., $[Hb_{diff}]$) was lower and $[HHb]$ higher than during RSN. Furthermore, in RSH-VHL, cerebral $[Hb_{diff}]$ was lower than baseline values during the large majority of the shuttle-run sprints. While systemic hypoxemia has been shown to play an important role in cerebral deoxygenation during exercise [1], the present results are original in the sense that, unlike an exposure to ambient hypoxia, VHL does not induce a continuous fall in SpO_2 . Therefore, a limited impact could have been expected when arterial desaturation is induced by VHL. Previously, earlier occurrence and larger extent in the prefrontal cortex deoxygenation have been reported during RSH at $FiO_2 \sim 13\%$ compared with normoxic conditions [4, 28, 34]. This demonstrates that, although dis-

continuous, the drop in SpO_2 during RSH-VHL is large enough to provoke a significant decrease in cerebral oxygenation, as shown by the strong correlation we found between SpO_2 and $\Delta[Hb_{diff}]$.

The fact that RSH-VHL did not induce a greater cerebral blood volume (i. e., higher $\Delta[THb]$) than in RSN represents an unexpected outcome. Indeed, hypoventilation leads to hypercapnia, which has long been known to augment cerebral vasodilation and therefore perfusion [14]. Even though cerebral $[THb]$ in RSH-VHL was higher than baseline values during the entire S2, it was not different from RSN. This may be partly explained by the fact that in the former condition, the $PETCO_2$ levels, and therefore hypercapnia, were not very strong. This was probably because the periods of sprints were very short whereas the following periods of recovery were 2.5 times longer.

While RSH-VHL elicited lower cerebral oxygenation, it is remarkable that the fatigue index was not greater than in RSN. Furthermore, the overall performance was not altered compared with RSN. This

finding is important because the effectiveness of the RSH method relies on the recruitment of a high number of fast-twitch fibers in order to induce physiological adaptations favorable to performance [12]. This result confirms what we recently reported in an RSH-VHL study in cycling [40] and may explain, at least in part, the large improvement in RSA reported after using this training method [16, 31]. Contrastingly, during RSH in normobaric hypoxia (i. e., altitude ≥ 3000 m; $\text{FiO}_2 < 14.5\%$), it has been shown that performance was not maintained compared with normoxic conditions [4, 7, 26]. Smith & Billaut [28] suggested that a lower cerebral oxygenation could play a role in the impaired RSA when exposed to hypoxia. Under these conditions, they found a decrease in surface electromyographic activity, and therefore in muscle activation, which was closely linked to the fall in cerebral oxygenation. They concluded that RSH may have a negative effect on cortical and subcortical functions of the central nervous system which could lead to what is termed central fatigue [9]. The occurrence of such phenomenon is, however, questionable in the present study, in particular, because unlike in RSH, the subjects breathe normoxic air (and without ventilatory pattern constraint) during the recovery periods following sprints with VHL. This makes SpO_2 rapidly return to normal levels (or close), which therefore favors the oxidative process of the recovery and consequently the maintenance in performance.

It is also of interest that $\dot{\text{V}}\text{O}_2$ was significantly higher in RSH-VHL than in RSN during the recovery periods. Such phenomenon has already been reported during both moderate [35] and repeated-sprint exercise [40] with VHL. It would be the consequence of the O_2 debt incurred during the breath hold periods [36, 37]. In the present study, the excess O_2 consumption during RSH-VHL could have been sufficient to preserve both muscle and cerebral O_2 delivery and consequently to enable the maintenance of performance compared with RSN. The inverse correlation we found between the change in the oxygen pulse, and to a lesser extent $\Delta\dot{\text{V}}\text{O}_2$, on the one hand, and SpO_2 and cerebral $\Delta[\text{Hb}_{\text{diff}}]$, on the other hand, suggests that the greater the fall in arterial and cerebral oxygenation, the higher the compensation in $\dot{\text{V}}\text{O}_2$. Considering that HR and muscle [HHb] were not different between RSH-VHL and RSN and that the oxygen pulse was significantly higher in the former condition, it could be concluded that the increase in was due to an elevated cardiac output, which would be the consequence of a higher stroke volume. An increased stroke volume was previously found during the recovery periods following VHL exercise [35]. It would be due to a “pump effect” as a consequence of the large and rapid inhalations that occur at the cessation of breath holding.

The higher $\dot{\text{V}}\text{O}_2$ during the recovery periods may also explain that $[\text{La}]_{\text{max}}$ tended to be lower at the end of RSH-VHL. So far, it has been clearly established that the glycolytic activity, and therefore lactate production, was increased during VHL exercise [36, 37]. In the present study, it is therefore highly likely that the 6-sec sprints with VHL elicited a greater stimulation of the anaerobic glycolysis. However, due to the higher levels of $\dot{\text{V}}\text{O}_2$ following the periods of sprints with RSH-VHL, it is also probable that lactate clearance was improved compared to RSN. Since the recovery periods were 2.5 times longer than the sprint periods, the balance may have shifted towards a greater lactate clearance and, as a result, a lower $[\text{La}]_{\text{max}}$ value, as recently reported in a cycling RSH-VHL study [40].

While the present study brings novel findings and confirms other recent outcomes, it has also several limitations. First, cerebral oxygenation was assessed only at the prefrontal cortex, which may not reflect the entire cerebral oxygenation. In addition, this area of the brain, unlike motor areas, is not directly involved in the neural control of movement. It is however important to note that prefrontal cortex deoxygenation has been suggested to contribute to fatigue during exercise [29]. Secondly, we did not perform any vascular occlusion in combination with NIRS to obtain quantitative measurements of muscle oxygen delivery and utilization. The problem with this technique is that it makes data comparison with other studies difficult due to the diversity of the methodologies used [3]. Conversely, analyzing the change in NIRS data relatively to resting baseline values is consistent with other RSH or RSH-VHL studies [4, 28, 40]. This makes comparisons easier. Third, it would have been useful to perform neuromuscular measurements, such as surface electromyography, to ascertain whether the lower cerebral oxygenation induced by RSH-VHL could alter the central motor command and impair voluntary muscle activation. It is however unlikely that this phenomenon happened in the present study since performance was not affected by this exercise intervention. Finally, it would also have been useful to non-invasively measure stroke volume and cardiac output to verify whether they were actually involved in the higher $\dot{\text{V}}\text{O}_2$ recorded during RSH-VHL.

In conclusion, this study showed that performing repeated shuttle-run sprints with VHL induced a greater drop in cerebral oxygenation compared to the same exercise with normal breathing. Despite this phenomenon, the fall in velocity throughout exercise was not greater in RSH-VHL than in RSN. The maintenance in performance in the former condition could be attributed to a higher $\dot{\text{V}}\text{O}_2$ during the recovery periods following sprints. Considering the large improvements in RSA that have been reported after training with RSH-VHL [16, 31], it would be interesting to investigate the physiological adaptations to this approach and determine to which extent they could be similar to those of RSH.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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